Reading & Interpreting Statistical Diagrams

Exam code: 0607

Download PDF

Written by: Mark Curtis
Reviewed by: Dan Finlay
Updated on 27 January 2026

Reading & interpreting statistical diagrams

How do I interpret statistical diagrams?

- Read and **understand** the initial sentences describing the **situation** (context)
  - **Underline** important words if necessary
- Look for any **keys** that may help you to understand the diagram
  - For example
    - 1 unit represents 20 people
    - Year 10 is the **solid** line, Year 11 is the **dotted** line
    - Class A is **shaded**, class B is **striped**
- **Read** the **titles** of diagrams and all **axes labels**
  - A graph for '**new** students at a school' is different to '**all** students at the school'
  - A **frequency** axis that **starts at 50** is different to one starting at 0
- Understand which **units** are being used
  - Individual lengths may be in **centimetres** but total length may be in **metres**
  - Populations may be measured in **thousands**
- Look out for any **extreme values** (outliers / anomalies)
  - One month's temperature might be unusually high
    - Was it a heat wave or a recording error?

How do I draw conclusions from diagrams?

- Look for **overall trends** in the diagram
  - Prices **increase** year on year
  - The temperature **peaks** in June
- Use **numbers** from the graphs
- Refer to any **changes**
  - The **steepness** (gradient) of graph may change
- Write in **full sentences** that copy the **exact wording** from the question

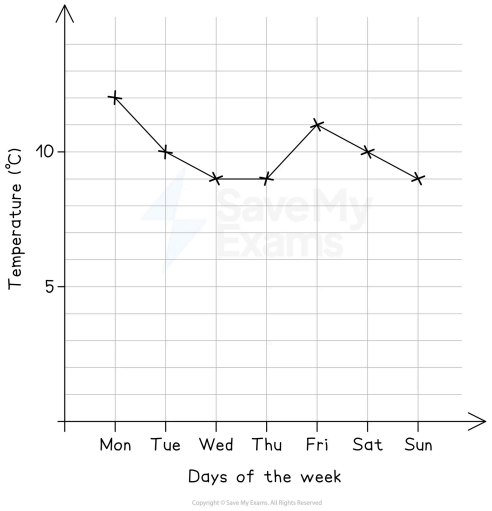
Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

- You may need to calculate the **mode**, **median**, **mean** or **range** to support any explanations
- Understand why drawing conclusions may **not** be **suitable**
  - The data set may be **too small** to be representative
  - The data set may be **biased**
  - Consider the **scope** of the data
    - e.g. Data for January to March cannot be used to predict August

Worked Example

The diagram below shows the temperature of a garden in the UK, recorded at 7am on each day of a particular week in March.



(a) Describe the change in temperature over the first four days.

Answer:

The trend shows a decrease in the first three days, then a constant temperature
Find numbers from the graph to use in your answer
Refer to the steepness of the change

The temperature decreases from 12°C on Sunday to 9°C on Wednesday
The decrease is steeper over the first two days
There is then a constant temperature of 9°C on both Wednesday and Thursday

(b) A gardener claims that, based on the graph, Monday must have experienced the highest temperature that week.

Give a reason as to why this might not be true.

Answer:

The temperatures on the graph are at 7am each day
The maximum temperature may have been after 7am, on a day that was not Monday

(c) A journalist wants to use the data shown to claim that the average temperature that week was below 10°C.
The mean of the temperatures shown is 10°C.

Which type of average would you suggest they use? Explain your answer.

Answer:

The mean of 10°C does not support the claim that the average temperature is below 10°C
Try calculating the mode instead

12, 10, 9, 9, 11, 10, 9

The most frequent number is 9

The modal temperature is 9°C
9°C < 10°C so using the mode would help the journalist's claim

Teacher only

Test your students on this in just a few clicks

- Create exam-aligned tests
- Tailor format & difficulty
- Quizzes or homework in minutes

Try Test Builder →

Test yourself

Was this revision note helpful? Yes No

Resources

Learning Hub

Scholarship

Join

Past Papers

Solution Banks

Sitemap

Dashboard

Overview

Brand ambassador

Profile

Subscription

Settings

Log out

Company

About us

Exam Specificity

Content Quality

Promotions

Jobs

Terms

Privacy

Cookie Policy

Help and Support

Subjects

Biology

Chemistry

Physics

Maths

Geography

English Literature

Psychology

All Subjects

© Copyright 2015–2026 Save My Exams Ltd. All Rights Reserved.

Neither IBO nor College Board were involved in the production of, and do not endorse, the resources created by Save My Exams.